

GORONTALO JALALUDDIN, Indonesia

WMO: 970480

Lat: 0.6385N

Lon: 122.8524E

Elev: 7

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	71.2	72.4	63.6	88.1	84.0	65.6	94.4	83.2	19.0	87.7	17.3	86.9	0.3	140	0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	13.4	92.7	76.9	91.7	77.1	90.8	77.3	80.5	88.4	79.9	87.7	79.5	87.1	9.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.3	147.0	83.3	77.9	144.9	82.9	77.4	142.4	82.5	44.1	88.6	43.5	88.4	43.1	87.8	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	13.5	11.6	64.6	95.8	5.1	2.0	60.9	97.2	57.9	98.3	55.1	99.4	51.4	100.8		
(5)			64.6	84.1	3.7	1.9	61.9	85.4	59.8	86.5	57.7	87.6	55.0	88.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.8	81.3	81.3	82.0	82.6	82.4	81.4	80.9	81.1	81.7	82.4	82.4	82.1
	DBStd	1.59	1.44	1.45	1.51	1.51	1.53	1.65	1.51	1.48	1.49	1.44	1.45	1.42
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11611	971	878	993	979	1004	942	957	965	950	1005	973	994
	CDD65	6136	506	458	528	529	539	492	492	500	500	540	523	529
	CDH74	59822	4744	4306	5235	5245	5371	4578	4579	4933	5160	5435	5134	5102
CDH80	21167	1483	1385	1845	1851	1896	1501	1532	1814	2132	2185	1839	1704	
Wind	WSAvg	3.0	2.9	3.1	3.1	2.6	2.3	2.5	3.6	4.7	3.8	2.8	2.5	2.4
Precipitation	PrecAvg	71.5	7.8	5.6	6.2	6.7	7.2	6.5	4.9	2.9	3.1	5.3	7.6	7.7
	PrecMax	109.3	19.8	14.0	19.3	12.4	18.0	13.3	10.7	11.5	13.7	13.0	19.8	16.9
	PrecMin	43.6	3.3	1.4	0.9	1.8	2.3	0.8	0.7	0.6	0.9	0.4	4.2	1.6
	PrecStd	17.3	3.1	3.6	4.5	2.1	3.4	3.5	3.1	2.8	2.8	3.8	3.4	3.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.9	90.8	92.4	93.1	91.8	90.9	90.8	91.6	93.4	93.7	93.4	92.0
		MCWB	78.8	77.8	77.6	78.4	79.2	78.7	78.0	76.3	75.7	75.9	77.1	79.0
	2%	DB	89.1	89.5	90.7	91.2	90.6	89.7	89.5	90.4	92.1	92.4	91.6	90.2
		MCWB	78.2	77.7	77.5	78.3	78.9	78.3	77.3	75.9	75.5	76.1	77.7	78.8
	5%	DB	88.1	88.3	89.4	89.8	89.6	88.5	88.4	89.3	90.8	91.1	89.9	89.0
		MCWB	78.0	77.5	77.4	78.3	78.7	78.0	77.0	75.6	75.4	76.3	78.0	78.6
	10%	DB	86.9	87.1	88.0	88.4	88.4	87.3	87.3	88.1	89.5	89.6	88.3	87.6
		MCWB	77.8	77.3	77.3	78.2	78.4	77.8	76.7	75.5	75.4	76.7	78.0	78.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.1	79.7	79.9	80.8	81.1	80.8	80.0	79.0	79.8	80.0	80.4	80.9
		MCDB	87.8	87.7	88.1	88.9	89.0	88.2	88.0	87.3	88.8	88.2	88.9	88.4
	2%	WB	79.2	78.9	79.0	79.9	80.1	79.8	79.0	78.0	78.4	79.1	79.6	79.9
		MCDB	86.7	86.9	87.1	87.7	87.7	87.1	86.8	86.4	87.1	87.4	87.6	87.4
	5%	WB	78.7	78.2	78.3	79.2	79.7	79.1	78.2	77.3	77.6	78.5	79.0	79.2
		MCDB	86.0	85.6	85.9	86.7	86.8	86.1	85.5	85.3	86.0	86.6	86.7	86.4
	10%	WB	78.1	77.7	77.8	78.7	79.1	78.4	77.6	76.5	77.0	77.9	78.4	78.7
		MCDB	84.7	84.6	85.0	85.8	85.7	84.9	84.4	84.1	85.0	85.4	85.5	85.6
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.7	13.5	13.4	13.4	12.9	13.4	14.6	16.7	16.1	14.0	12.8
		MCDBR	13.7	14.2	15.0	14.8	14.8	14.6	15.2	16.6	18.8	18.2	15.8	14.2
		MCWBR	4.5	4.5	4.4	4.3	4.7	5.1	5.1	5.0	5.6	5.4	4.7	4.5
	5% WB	MCDBR	12.9	13.1	13.5	13.7	13.6	13.1	13.6	14.2	15.7	15.1	14.0	13.3
		MCWBR	4.8	4.6	4.5	4.6	4.9	5.0	5.2	5.0	5.5	5.0	4.8	4.8
Clear-Sky Solar Irradiance	taub		0.426	0.428	0.438	0.435	0.425	0.428	0.424	0.440	0.455	0.503	0.457	0.435
	taud		2.438	2.426	2.394	2.402	2.446	2.453	2.468	2.403	2.347	2.190	2.343	2.421
	Ebn at Noon		287	287	282	277	272	267	270	272	273	263	275	282
	Edh at Noon		38	39	40	38	36	35	35	38	41	49	41	38
All-Sky Solar Radiation	RadAvg		1611	1694	1766	1769	1685	1526	1611	1730	1866	1820	1705	1611
	RadStd		77	108	136	88	68	89	147	122	128	82	80	101

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page